Adaptive Multi-Agent Digital Twin Framework for Speech-Enabled Explainable Clinical Decision Support

An AI Systems Research Project — Healthcare as Application Domain

Submitted to: Prof. M. P. S. Bhatia | B.Tech Final Year Project Proposal

VISION

Existing healthcare AI assistants are single-model chatbots: one LLM, one prompt, one answer, with no memory of how the patient's condition is evolving, no mechanism to bring in specialist perspective, and no stated confidence in what it just told you. This project is not another instance of that pattern. It is an AI-systems project that uses clinical decision support as its test domain — the contribution is an adaptive multi-agent architecture with a speech-first interface, structured explainability, and calibrated uncertainty, evaluated in a healthcare setting.

PROBLEM STATEMENT

- **Static records.** Electronic Health Records store history but cannot simulate a future state or compare hypothetical interventions.
- **Monolithic reasoning.** A single LLM answering every query cannot represent specialist disagreement, cannot decide when a second opinion is needed, and offers no principled way to combine conflicting expert views.
- **No accessible input channel.** Text-only interfaces exclude elderly, visually-impaired, low-literacy, and rural multilingual users — exactly the populations most in need of decision support.
- **Overconfident, opaque output.** Most systems return an answer with no reasoning trail, no cited evidence, and no signal of how confident the model actually is.

This project addresses the gap between AI chatbots and AI systems: architectures that route, simulate, reconcile, and explain, rather than simply complete a prompt.

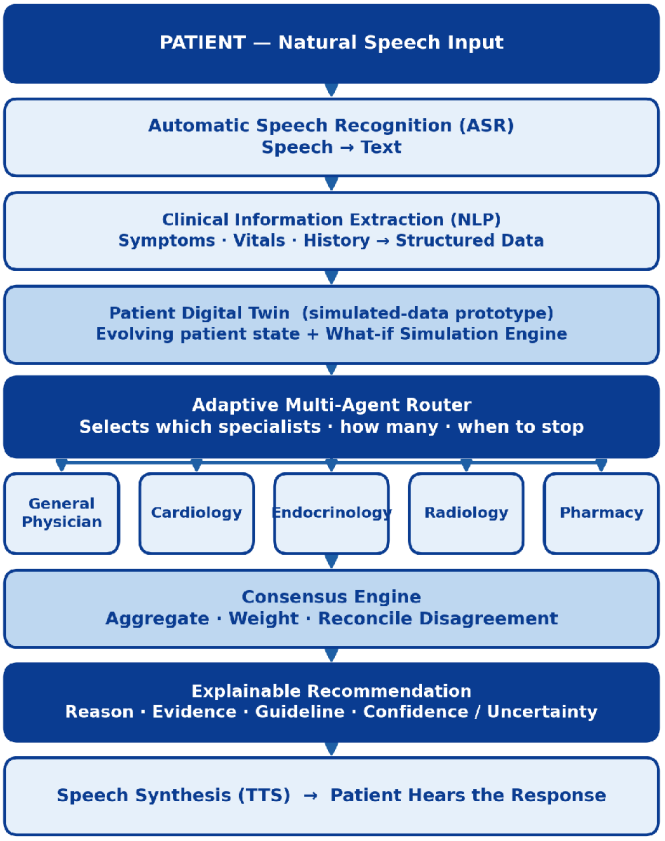
OBJECTIVES

- Design and implement an adaptive multi-agent router that selects which specialist agents to invoke, and when to stop, based on symptom pattern and model uncertainty.
- Build a consensus mechanism that aggregates and reconciles disagreeing specialist recommendations into a single weighted output.
- Implement a speech-in / speech-out interface (ASR → structured extraction → TTS) as the primary accessibility-oriented interaction mode.
- Prototype a lightweight, simulated-data patient state model (“digital twin”) to support illustrative what-if comparisons across treatment options.
- Attach evidence, guideline references, and calibrated confidence to every recommendation the system produces.
- Evaluate the multi-agent architecture against a single-LLM baseline on routing accuracy, response relevance, and explanation quality.

RESEARCH QUESTIONS

- Does adaptive multi-specialist routing produce more accurate and relevant recommendations than a single monolithic LLM, at comparable inference cost?
- How should conflicting recommendations from independent specialist agents be reconciled into one explainable output?
- Does a speech-first interface measurably improve task completion and usability for elderly and low-literacy users, versus text input?
- Can uncertainty be estimated in a way that is calibrated — i.e. the stated confidence actually tracks correctness — rather than a cosmetic number?

SYSTEM ARCHITECTURE



CORE RESEARCH COMPONENTS

Adaptive Multi-Agent Router Learns which specialist agents to invoke, how many, and when to stop, based on symptom pattern and uncertainty — the core AI-systems contribution.	**Consensus Engine** Aggregates and reconciles disagreeing specialist outputs (e.g. cardiology vs. nephrology) into one weighted, reasoned recommendation.
Speech Interface (ASR + TTS) Natural spoken interaction, positioned as an accessibility layer for elderly, visually-impaired, rural and multilingual Indian users.	**Simulated Patient Digital Twin** A lightweight, continuously-updated patient state model used for illustrative what-if comparisons — explicitly scoped as simulated-data, not clinically validated.
Explainability Layer Every recommendation is returned with reason, supporting evidence, guideline reference and specialist attribution — not a bare answer.	**Uncertainty Estimation** Recommendations are returned with a confidence estimate rather than presented as certain, flagging low-confidence cases for human review.

SCOPED DELIVERY PLAN

The full architecture above is the target system. What is realistically buildable and defensible within a B.Tech timeline is delivered in three phases, with Phase I and II as the committed deliverable and Phase III as the stretch target:

Phase	Focus	Deliverable
I	Speech-to-structured-data pipeline (ASR + clinical NLP extraction); baseline single-LLM assistant for comparison	Working ASR → structured-symptom module; baseline system
II	Adaptive multi-agent router + specialist agents (GP, Cardiology, Endocrinology, Pharmacy) + Consensus Engine	Multi-agent prototype outperforming baseline on routing accuracy / relevance
III	Simulated-data Digital Twin module + uncertainty-calibrated, evidence-linked explanations; TTS output; end-to-end evaluation	Full pipeline demo + evaluation report + paper draft

EXPECTED OUTCOMES

- A working end-to-end prototype: speech input → multi-agent reasoning → explained, spoken recommendation.
- A quantitative comparison of adaptive multi-agent routing against a single-LLM baseline.
- An open-source reference implementation of the router, consensus engine, and explainability layer.
- A written evaluation report suitable for adaptation into a conference-length paper (systems / applied-AI track).

RELEVANCE

Adaptive agent routing, multi-agent collaboration, and explainable, uncertainty-aware AI systems are active research directions across the field — not unique to this proposal, but a genuine and current systems problem worth undergraduate-level investigation, with clinical decision support as one of several domains where the architecture can be evaluated.